Week 11 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred (*) proof problems are eligible for inclusion in the proof portfolio assignment.

11.1 Exercises

11.1.1 Exercise

From the text, Exercise 3.6.

11.1.2 Exercise

From the text, Exercise 3.8.

11.1.3 Exercise

From the text, Exercise 3.9.

11.1.4 Exercise

Let G be a connected 3-regular planar graph on 8 vertices. How many regions does G have? Give an example of a 3-regular connected graph on 8 vertices that is not planar or explain why it must be planar.

11.1.5 Exercise

For the following degree sequences (list of the degrees of the vertices in a graph) determine if there exists a bipartite simple graph with these degrees. If yes construct it, if not explain why it does not exist.
a. (6, 6, 4, 4, 4, 4, 4, 4, 2, 2, 2)
b. (4, 4, 4, 4, 4, 3, 3, 2, 2)
c. (5, 5, 5, 5, 5, 5, 4, 4, 4)

11.1.6 Exercise

What is the smallest number of colors that can be used to color the vertices of a cube so that no two adjacent vertices are colored identically?

11.1.7 Exercise

Euler's formula $v - e + f = 2$ holds for all connected simple planar graphs. What if a graph is not connected? Suppose a planar graph has two components. What is the value of v-e+f now? What if it has k components?

11.2 Proofs

11.2.1 Proof

Prove the chromatic number of any tree is two. Recall, a tree is a connected simple graph with no cycles.

11.2.2 Proof

Prove that if each component of a simple graph is bipartite, then the entire graph is bipartite.'

11.2.3 Proof

Let G be a connected simple graph on n vertices. Prove that if $n \geq 11$ then either G or the complement of G is not planar.

11.2.4 Proof

Prove that if a simple connected planar graph has e edges and v vertices with $v \geq 3$ and no cycles of length 3 then $e \leq 2v - 4$.

11.2.5 Proof

Prove Euler's formula using induction on the number of **edges** in the graph.

11.2.6 Proof

Prove G is regular if and only if it's complement is regular.

11.2.7 Proof *

Prove that any simple connected planar graph must have a vertex of degree 5 or less.

11.2.8 Proof *

Prove that a simple graph with 17 vertices and 73 edges cannot be bipartite. (Hint: Break the graph up into two vertex sets and count the number of edges within each vertex set.)

11.2.9 Proof *

Prove Dirac's Theorem: If G is a simple graph with p vertices, each vertex having degree $\geq \frac{1}{2}p$, then G is hamiltonian. Hint: Suppose G is not hamiltonian but adding any one edge will make G hamiltonian, so G has a path $v_1 \to v_2 \to \cdots \to v_p$ containing all the vertices of G. Show there must be a vertex adjacent to both v_1 and v_p, and use this to create a hamiltonian cycle.

11.2.10 Proof *

Let G be a graph (not necessarily connected as in a problem above) on n vertices. Prove that if $n \geq 11$ then either G or the complement of G is not planar.